Delta cell CAD v2: drawing set

All dimensions in mm. Arm parts are drawn in the arm-local frame: origin on the arm axis at radius R = 110, x radial outward, y along the arm axis, z up. Rod ends are drawn in the rod frame (ball centre at the origin, rod along +z'). General tolerance for printed parts ISO 2768-m (+-0.2); fits and critical dimensions are in each sheet's table. Hidden lines are grey.

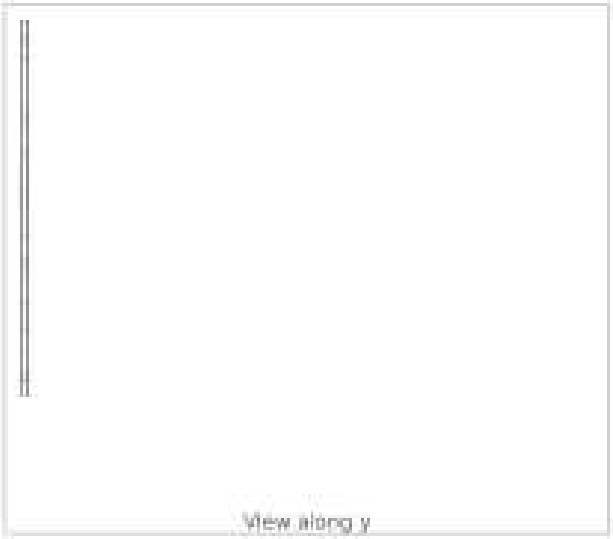
Axial stack of one arm module (along the arm axis, y)

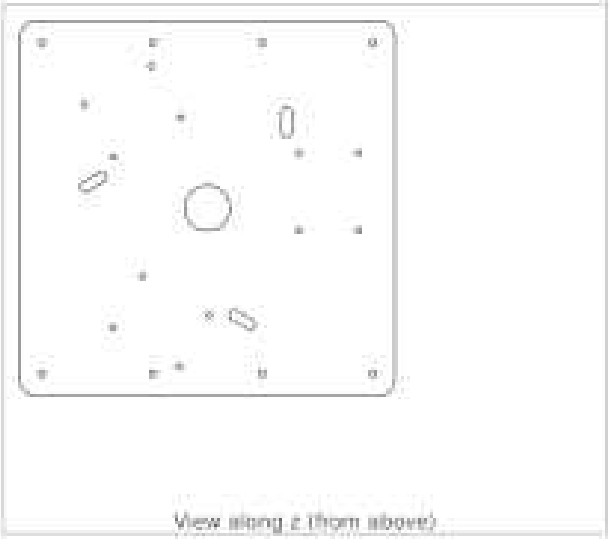
Component	y from	y to	length
Encoder board face	-45.0	-45.0	0.0
Magnet	-43.5	-41.0	2.5
Magnet holder	-43.5	-34.0	9.5
Shaft collar 8 x 16 x 8	-34.0	-26.0	8.0
Spacer 12 x 8.2 x 2.5	-26.0	-23.5	2.5
Bearing 608-2RS (a)	-23.5	-16.5	7.0
Cheek (a)	-25.0	-15.0	10.0
Spacer 12 x 8.2 x 8.5	-16.5	-8.0	8.5
Arm hub	-8.0	8.0	16.0
Spacer 12 x 8.2 x 8.5	8.0	16.5	8.5
Cheek (b)	15.0	25.0	10.0
Bearing 608-2RS (b)	16.5	23.5	7.0
Spacer 12 x 8.2 x 3.0	23.5	26.5	3.0
120T pulley hub	26.5	46.0	19.5
Belt teeth zone (both pulleys)	34.0	45.0	11.0
20T pulley hub	46.0	53.0	7.0
Motor plate	53.5	57.5	4.0
Motor face	57.5	57.5	0.0
Shaft 8 h6 x 84	-38.0	46.0	84.0

Assembly: press both bearings into the cheeks from the inside; slide the shaft from +y through bearing b, spacer, hub, spacer, bearing a; set the 3.0 spacer and the 120T pulley on +y, the 2.5 spacer and the collar on -y; preload lightly with the collar, tighten the pulley set screws on the flat, then the hub clamp. Glue the magnet holder last, with the board removed. Belt: 10 mm GT2, 300 mm closed (150 teeth), centre distance 72.94, tension by sliding the motor in its slots.

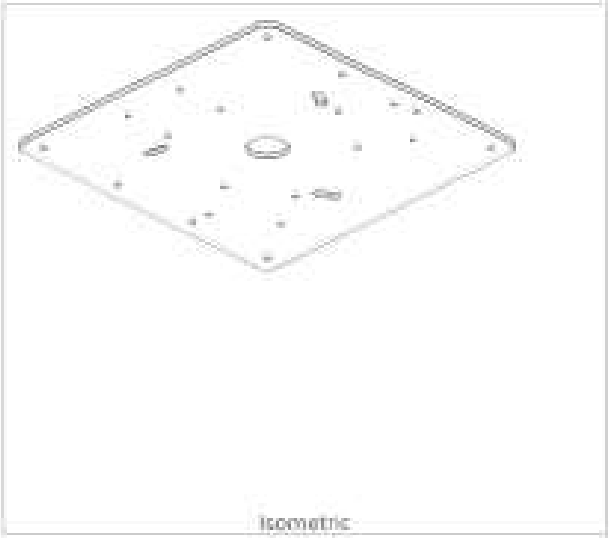
Project	High-speed delta cell, 4th SC
Sheet	0 / 10: stack
Source	cad/delta_cad (generated from config/robot.yaml)

M01 Base plate







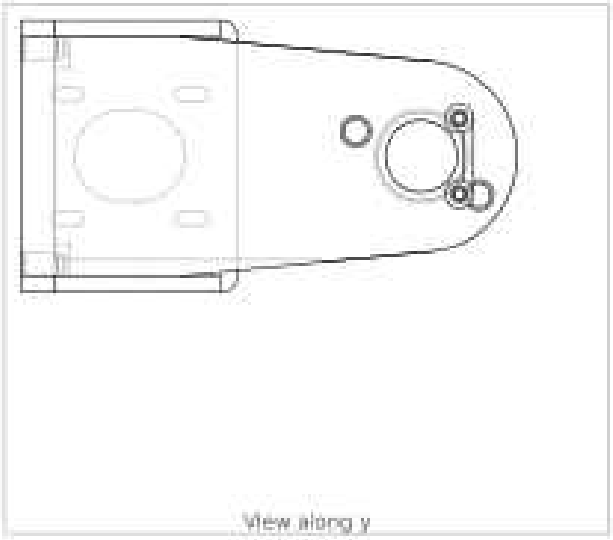


Feature	Dimension	Tolerance / note
Outline	340.0 x 340.0 x 6.0, corners R15, edges 0.5 x 45	+-0.2
Flatness	0.10 over the plate	check on a granite plate
Centre hole	dia 40 (vacuum tube, cables)	+-0.2
Bracket holes	12 x dia 4.5 through, 4 per arm at arm-local (+-27, -20) and (+-27, 50) about (R=110, phi)	position +-0.05
Frame holes	8 x dia 5.5 at x = +-50, +-150 and y = +-150 (M5 into T-nuts of the 2020 rails)	+-0.1
Cable slots	3 x 26 x 9 at arm-local (-38, 78)	
Envelope	340.0 x 340.0 x 6.0	x, y, z extents

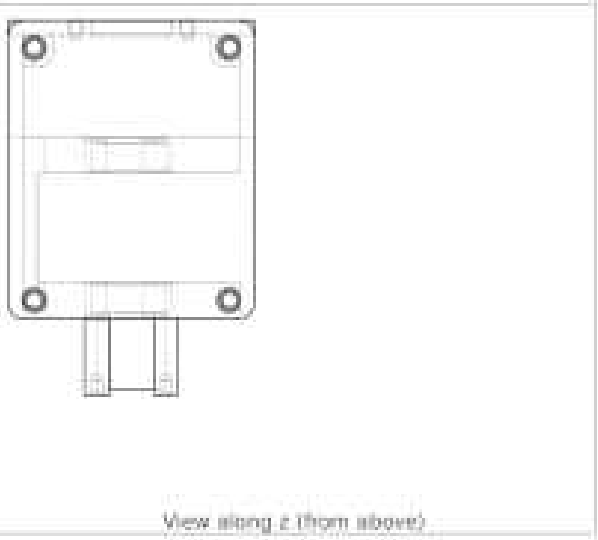
Material: 6061-T6 aluminium, 6 mm. STEP and STL: cad/out/parts/M01_*.step.

Part	M01 Base plate
Sheet	1 / 10
Units	mm, frame as in sheet 0

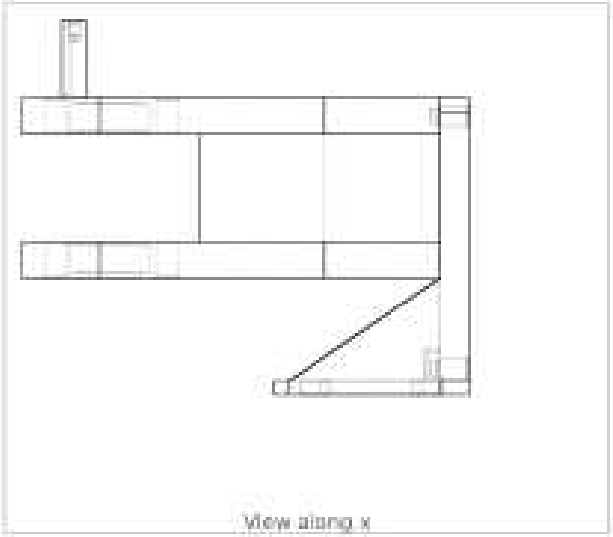
M02 Pivot bracket with motor plate (x3)



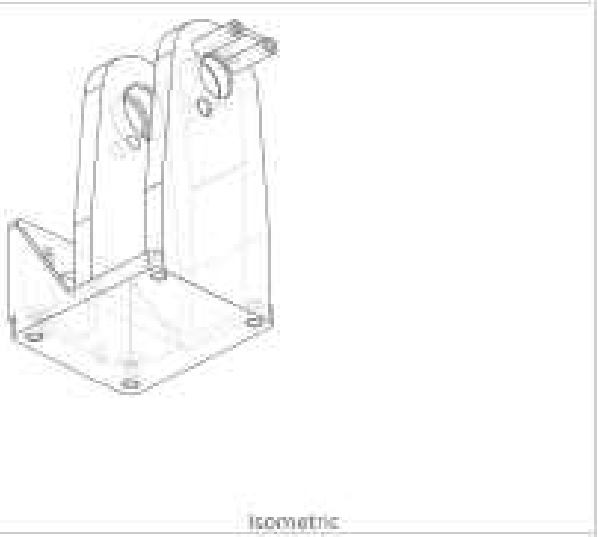
View along y



View along z (from above)



View along x



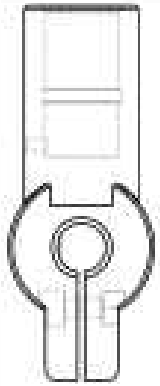
Isometric

Feature	Dimension	Tolerance / note
Arm axis to flange top	100.00	+0 / +0.10: sets R and the arm height
Cheeks	10 thick at |y| 15..25, gap 30.0	+-0.1
Bearing seats (608)	dia 22.00, depth 8.5 from the inner face, lead-in 0.6 x 45, shoulder dia 18	print a coupon: target light press, 0 to -0.02
Bearing seats coaxial	both seats on one axis	0.05: ream both from one side with a 22 mm reamer
Hard-stop pin holes	2 x dia 6.0 H7 through both cheeks at (x, z) = (-9.79, -14.50), (6.26, 16.33)	drill 5.8, ream 6.0
Motor plate	4.0 thick at y 53.5..57.5, boss slot 22.6 x +-2.5	
Motor screws	4 x slot 3.4 x +-2.5 on 31.0 square about the motor axis	
Motor axis above arm axis	72.94 nominal (300 mm GT2 belt), adjustable +-2.5	set by belt tension
Flange inserts	4 x M4 heat-set (hole dia 5.6, depth 8.6) from the top face	
Encoder posts	2 x dia 7 to y = -46.5, M2.5 inserts (hole 3.2)	board-to-magnet gap 1.5
Envelope	68.0 x 104.0 x 124.0	x, y, z extents

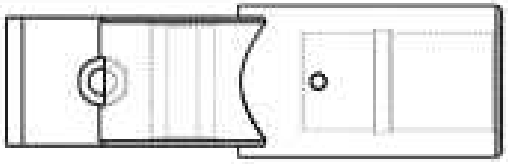
Material: PETG-CF, 5 walls, 40 % gyroid, print flange-down. STEP and STL: cad/M02_pivot_bracket.*

Part	M02 Pivot bracket with motor plate (x3)
Sheet	2 / 10
Units	mm, frame as in sheet 0

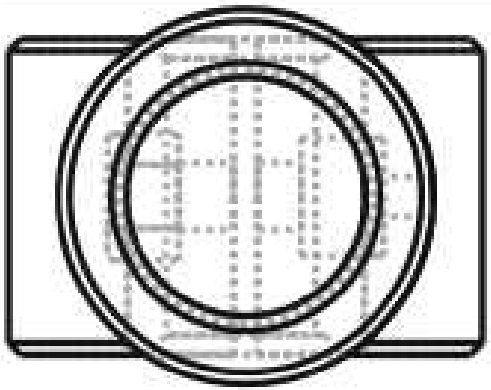
M04 Arm hub (x3)



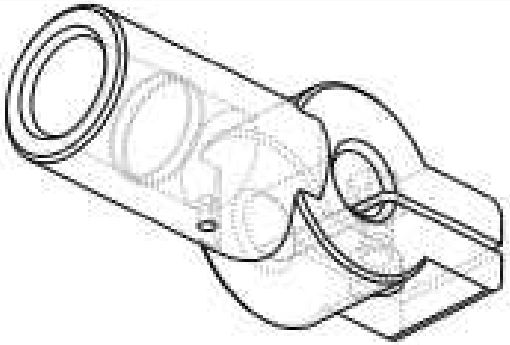
View along y



View along z (from above)



View along x



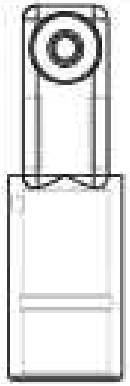
Isometric

Feature	Dimension	Tolerance / note
Width	16.0 (y -8..8)	+-0.05; spacers take up the rest
Shaft bore	dia 8.05 with D-flat at 3.55 from the axis	slide fit on 8 h6
Clamp	slit 1.2 through the tail, M3 x 10 at x = -10, counterbore dia 6, nut trap 5.6 AF	
Tube socket	dia 12.10, x 15..40 (bond length 25), lead-in 0.6 x 45, glue groove at x 24..26	0.05 radial glue gap; epoxy (Araldite 2011 or similar)
Vent	dia 2 from the socket bottom to the top	lets air and glue escape
Hard-stop tail	to x = -22, faces at z = +-6	touches the pins at -25 and +80 deg
Envelope	62.0 x 19.0 x 24.0	x, y, z extents

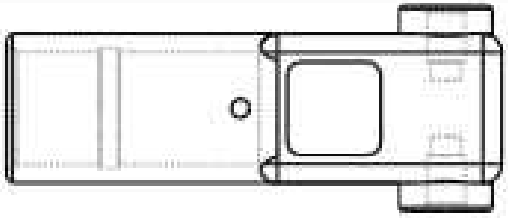
Material: PETG-CF, 100 % infill within 6 mm of the bores. STEP and STL: cad/out/parts/M04_*.step.

Part	M04 Arm hub (x3)
Sheet	3 / 10
Units	mm, frame as in sheet 0

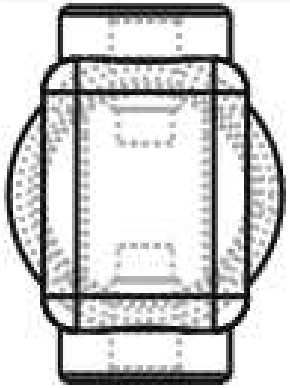
M05 Elbow block (x3)



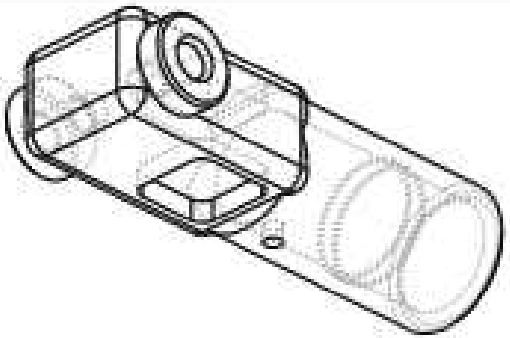
View along y



View along z (from above)



View along x



Isometric

Feature	Dimension	Tolerance / note
Tube socket	dia 12.10, x 123..150 (bond 27)	coaxial with the hub socket via the bonding jig
Ball-stud axis	x = 170.0 from the arm axis (sets L1)	+-0.10 after bonding (jig)
Stud faces	y = +-11.0; ball centres at y = +-20.0 (w = 40)	+-0.05
Inserts	2 x M3 heat-set, hole dia 4.0 x 6.0, then dia 3.2 to y = +-3.0	stud thread M3 x 6 + Loctite 243
Block	x 150..176, 22 x 12, R2 edges, 2 pockets 10 x 14 x 2.5	
Envelope	53.0 x 22.0 x 16.0	x, y, z extents

Material: PETG-CF. STEP and STL: cad/out/parts/M05_*.step.

Part	M05 Elbow block (x3)
Sheet	4 / 10
Units	mm, frame as in sheet 0

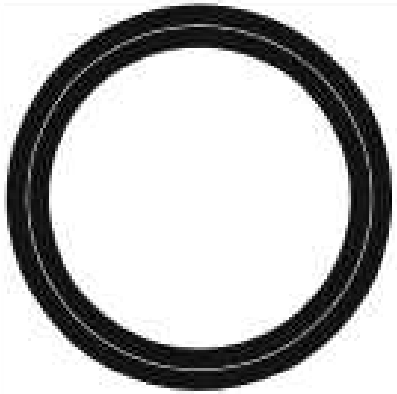
M06 Upper arm tube (x3)



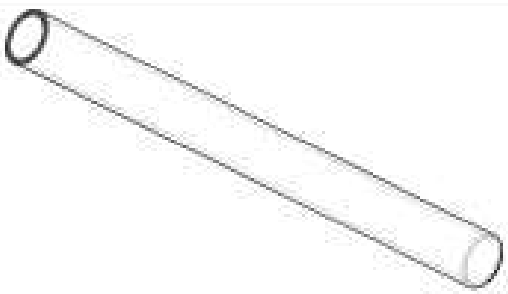
View along y



View along z (from above)



View along x



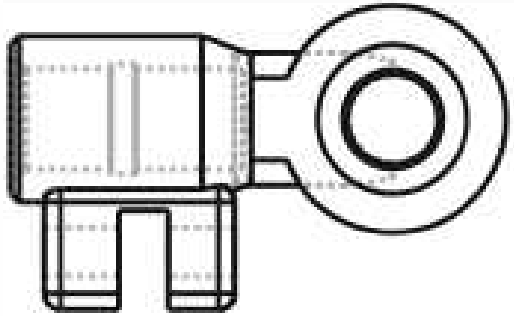
Isometric

Feature	Dimension	Tolerance / note
Length	135.0	+-0.2, ends square and deburred
Bonding	abrade the ends 25 / 27 mm, clean with IPA	
Envelope	135.0 x 12.0 x 12.0	x, y, z extents

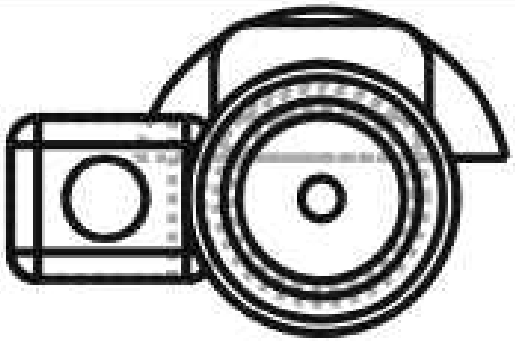
Material: pultruded carbon tube 12 x 10. STEP and STL: cad/out/parts/M06_*.step.

Part	M06 Upper arm tube (x3)
Sheet	5 / 10
Units	mm, frame as in sheet 0

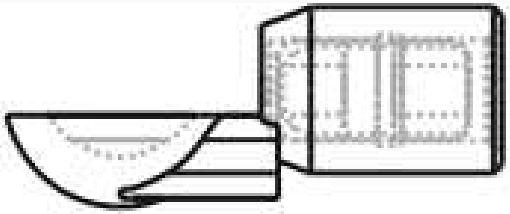
M07 Rod end (x12: 6 as drawn, 6 mirrored)



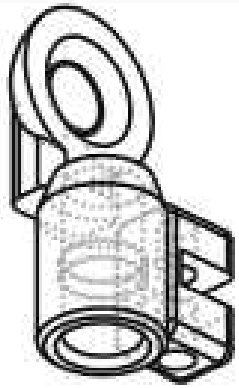
View along y



View along z (from above)



View along x



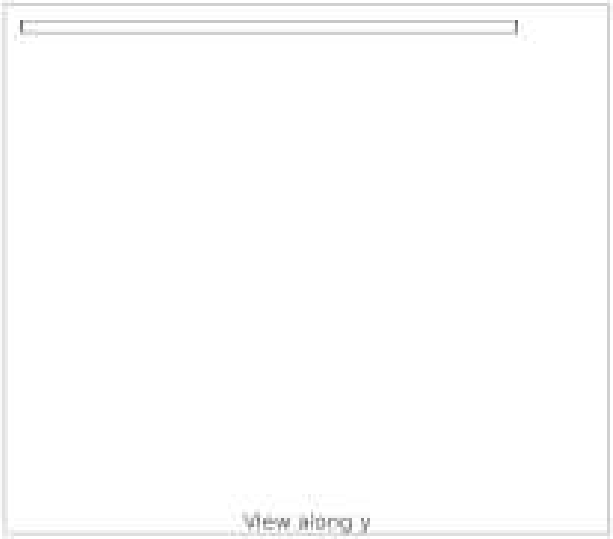
Isometric

Feature	Dimension	Tolerance / note
Seat	sphere dia 8.1 about the ball centre; rim plane 1.5 past the centre	test print with a ball first
Socket	dia 6.1, z' 9..23 (bond 14), groove at 15.5..17	
Spring pin	3 x 12 dowel, hole dia 2.95, at x' = -8; hook slot 3 wide at z' = 15	press fit
Swing	cup rim clears the stud neck up to 123 deg between rod and stud axis	checked in the CAD at 20 poses
Envelope	18.3 x 12.0 x 29.8	x, y, z extents

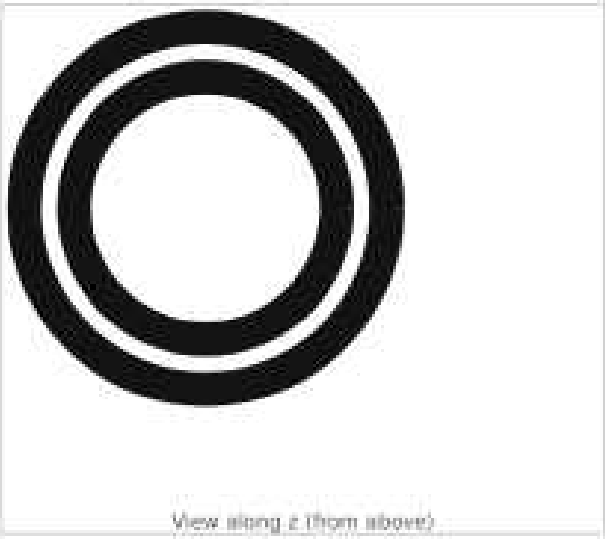
Material: PETG-CF, seat up, 0.12 mm layers. STEP and STL: cad/out/parts/M07_*.step.

Part	M07 Rod end (x12: 6 as drawn, 6 mirrored)
Sheet	6 / 10
Units	mm, frame as in sheet 0

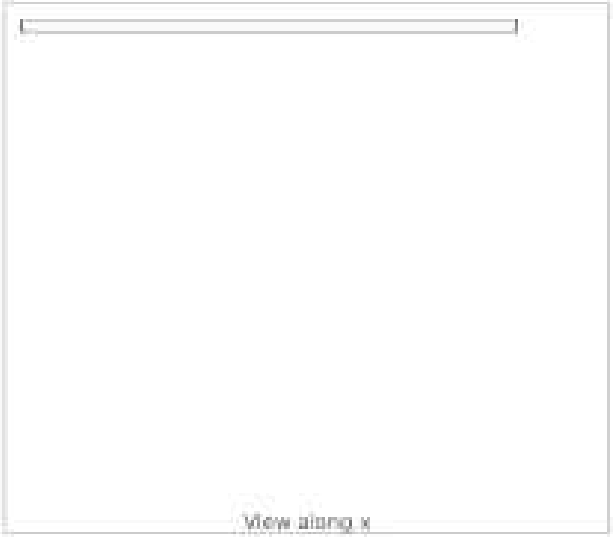
M08 Rod tube (x6)



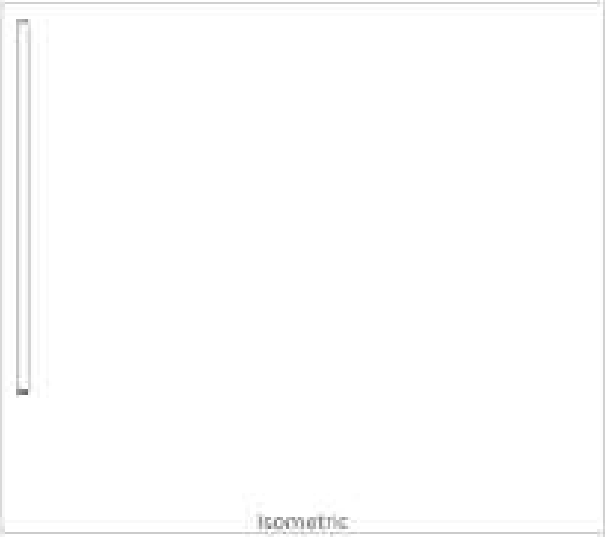
View along y



View along z (from above)



View along x



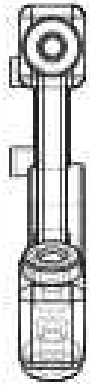
Isometric

Feature	Dimension	Tolerance / note
Length	242.0	+-0.2
Ball-to-ball after bonding	260.00	+-0.05, all six equal within 0.05: bond in a jig with two 8 mm balls
Envelope	6.0 x 6.0 x 242.0	x, y, z extents

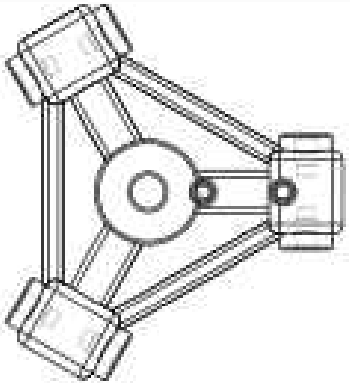
Material: pultruded carbon tube 6 x 4. STEP and STL: cad/out/parts/M08_*.step.

Part	M08 Rod tube (x6)
Sheet	7 / 10
Units	mm, frame as in sheet 0

M09 Platform



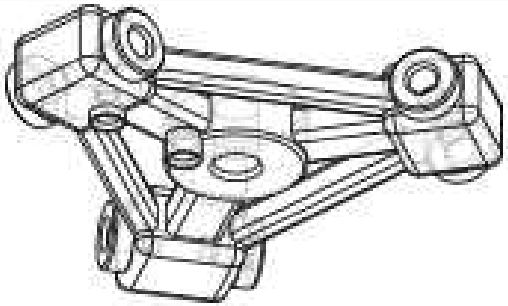
View along y



View along z (from above)



View along x



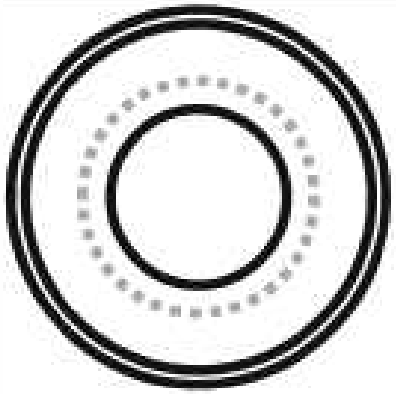
Isometric

Feature	Dimension	Tolerance / note
Ball centres	pairs 40 apart at radius 30, at 0 / 120 / 240 deg, all in z = 0	+-0.05
Stud faces	t = +-11 on each lug	
Inserts	6 x M3 (hole 4.0), 1 x M5 through the hub (hole 6.4), 2 x M2.5 for the ADXL345	
Cup	M5 cup holder from below, push-in fitting from above; lip at z = -35 (TCP)	
Envelope	63.8 x 71.2 x 12.5	x, y, z extents

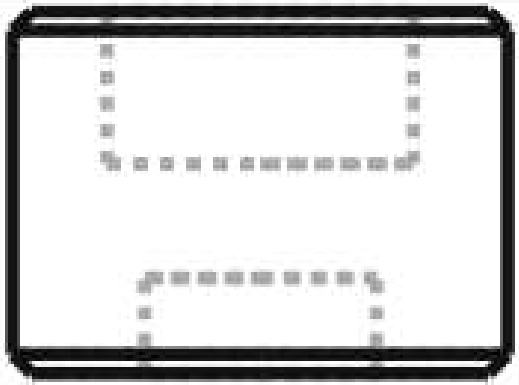
Material: PETG-CF, print flat. STEP and STL: cad/out/parts/M09_*.step.

Part	M09 Platform
Sheet	8 / 10
Units	mm, frame as in sheet 0

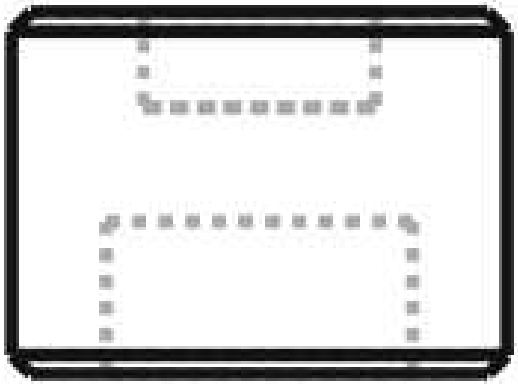
M10 Magnet holder (x3)



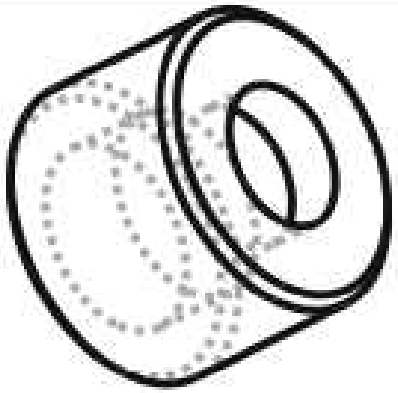
View along y



View along z (from above)



View along x



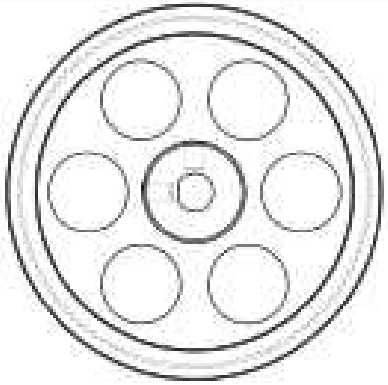
Isometric

Feature	Dimension	Tolerance / note
Bore	dia 8.05 x 4 on the shaft end	glue
Magnet pocket	dia 6.1 x 2.5 for a 6 x 2.5 diametric magnet, 3.0 from the steel shaft	
Envelope	13.0 x 9.5 x 13.0	x, y, z extents

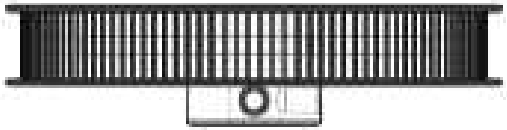
Material: PLA (non-magnetic). STEP and STL: cad/out/parts/M10_*.step.

Part	M10 Magnet holder (x3)
Sheet	9 / 10
Units	mm, frame as in sheet 0

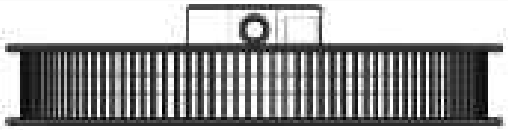
M11 Pulley GT2 120T, 10 mm belt (x3)



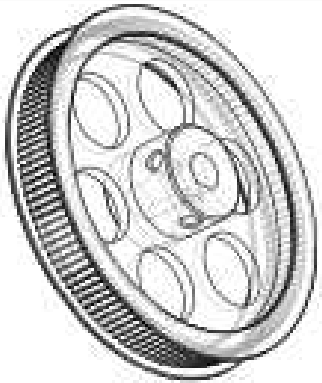
View along y



View along z (from above)



View along x



Isometric

Feature	Dimension	Tolerance / note
Teeth	120 x GT2, PD 76.394, OD 75.886, tooth depth 0.764, width 1.494 at the land	print a 20-tooth sector and test with the belt
Tooth width	11.0 (y 34..45), flanges dia 80.9 x 1.0	
Hub	dia 22, y 26.5..46.0, D-bore 8.05, 2 x M3 inserts at y = 30.5 (on the flat and at 90 deg)	
Alternative	buy an aluminium GT2 120T 8 mm bore pulley with the same tooth zone	
Envelope	80.9 x 19.5 x 80.9	x, y, z extents

Material: PETG-CF, 0.12 mm layers, flat. STEP and STL: cad/out/parts/M11_*.step

Part	M11 Pulley GT2 120T, 10 mm belt (x3)
Sheet	10 / 10
Units	mm, frame as in sheet 0